

QIYU DONG

BSc Mathematics (Pure Mathematics Track), HKUST | Algorithms · Optimization · Machine Learning
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EDUCATION

Hong Kong University of Science and Technology (HKUST) — BSc Mathematics, Pure Mathematics Track

Sep 2024 – Present

CGA: 3.436/4.3. Selected grades: Discrete Structures A+; Combinatorial Analysis A; Honors Linear & Abstract Algebra I/II A; Introduction to Lie Algebras A; Programming with C++ A; General Biology I/II A.

National University of Singapore (NUS) — Exchange Student

Aug – Dec 2026

RESEARCH & TECHNICAL PROJECTS

3D Container Loading Optimization | C++, Python, OR-Tools CP-SAT, Reinforcement Learning

Independent project

- Built C++ heuristic and recursive-search methods for 3D packing with geometric feasibility checks and six box orientations.
- Formulated a CP-SAT model in Python with item selection, rotation, placement coordinates, and pairwise non-overlap constraints.
- Developed a DQN-style sequential packing approach using height-map state features, candidate actions, replay, target networks, and reward shaping; current work explores reproducible benchmarks, candidate ranking, and warm starts.

Computational Knot Invariants & Quantum Groups | Python, SymPy

Research / UROP-related work

- Implemented and verified $U_q(\mathfrak{sl}_2)$ R-matrix computations, including Yang–Baxter equation checks, braid eigenvalues, and minimal-polynomial calculations.
- Extended experiments to $U_q(\mathfrak{sl}_3)$ fundamental and $\mathfrak{sl}_2$ spin-1 local data; studied tensor-product decompositions, channel projectors, and a modular pipeline from algebraic input to knot/link invariants.

SELECTED COURSEWORK

Mathematics: Honors Probability, Honors Mathematical Analysis, Honors Linear & Abstract Algebra I–II, Combinatorial Analysis, Discrete Structures, Lie Algebras.

Computing / AI: Programming with C++, Object-Oriented Programming & Data Structures, Exploring Artificial Intelligence, Mathematics for AI.

Life Science: General Biology I & II; Laboratory for General Biology I.

TECHNICAL SKILLS

Programming: C++, Python, Java | **Tools / Libraries:** OR-Tools CP-SAT, SymPy, Git/GitHub | **Methods:** combinatorial optimization, heuristic search, mathematical modeling, reinforcement-learning fundamentals

SELECTED ACHIEVEMENT

China Thinks Big Project — Global Finalist; National Second Prize (Data Analyst)